Independent Program Verification Outperforms Translation in Diagnosing Reasoning Errors in Small Language Models: A Causal and Metacognitive Framework

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Abstract

Chain-of-thought (CoT) prompting frequently suffers from post-hoc rationalization, wherein intermediate verbalized rationales fail to causally govern model answers or merely mimic flawed heuristic assumptions. In this paper, we investigate whether deterministic programmatic verifiers and structured reasoning representations can reliably diagnose reasoning fallacies and execution slips in fine-tuned 8-billion parameter models (Qwen3-8B 4-bit with LoRA) without human intervention. We establish a pre-registered experimental evaluation across 500 challenging problems from the MATH dataset (Levels 3–5) and GSM8K. On GSM8K, structured `Step/Calc` fine-tuning achieves 95/100–97/100 zero-prefill format validity (passing Gate 0), high causal faithfulness under intermediate arithmetic perturbations (89.7%–93.8% causal following, with only 2.8%–3.1% reverting to unperturbed beliefs), and recovers 68.4%–78.9% (13/19 to 15/19) of legacy failure targets via Step 1 semantic enrichment (restoring 87/100 zero-prefill accuracy under `enriched_cf`, McNemar $p = 0.1094$ vs. `base_prefill`). However, scaling flat structured traces to symbolic mathematics reveals a fundamental structural bottleneck: internal equational verification is insufficient, catching only 2.4% (2/82) of erroneous solutions on MATH Levels 3–5 (items 101–300 under `base_showcalc`, false alarm rate 0/118), and although non-truncated LoRA traces achieve 72.7% (40/55) accuracy, paragraph-collapsed proofs induce greedy repetition loops in 75.6% (34/45) of truncated sequences. In one pre-registered development run on MATH (items 301–500; two automated labelling passes over 39 solutions, $\kappa = 0.947$, no human review), the results are consistent with the **Circular Confirmation Trap**: **independent programmatic generation catches 5× more reasoning errors than literal thought-trace translation** (10/24 [41.7%] vs. 2/24 [8.3%], exact two-sided McNemar $p = 0.0215 < 0.05$), while translation achieves parity on pure execution arithmetic (69.2% vs. 61.5%). To resolve the flat-trace repetition versus monolithic-interpreter crash trade-off on symbolic mathematics, we outline an encapsulated Directed Acyclic Graph (DAG) and selective tool-delegation architecture (MSRA) as future work.

1 Introduction

Modern autoregressive large language models demonstrate competitive performance when generating natural language rationales step by step [Wei et al., 2022, Cobbe et al., 2021]. Despite these empirical gains, extensive evidence reveals that verbalized rationales often function as post-hoc justifications rather than faithful computational engines [Lanham et al., 2023, Turpin et al., 2024]. In complex mathematical problem solving, reasoning errors fall into two fundamentally distinct categories:

1. **Execution Slips:** Arithmetic miscalculations where the underlying logical formula or conceptual relationship is correct, but the discrete evaluation is erroneous (e.g., $14 \times 7 = 96$).

2. **Reasoning Fallacies:** Conceptual formulation breakdowns where the model asserts an invalid mathematical premise, ignores physical constraints, or constructs a fundamentally incorrect equation (e.g., equating perimeter directly to area).

To mitigate arithmetic failure modes, prior works such as Program-Aided Language Models (PAL) [Gao et al., 2023] and Program of Thoughts (PoT) [Chen et al., 2023] outsource arithmetic evaluation to an external Python interpreter. When applying programmatic verification to monitor autoregressive generation, two divergent paradigms emerge:

- **Translated Trace Verification ($\mathcal{T}$):** The model translates its natural language thought trace into Python/SymPy code, which is then executed to verify whether the textually claimed answer matches the executed value.
- **Independent Program Verification ($\mathcal{I}$):** An external solver script is generated directly and independently from the problem text Q, bypassing the internal textual rationale.

A central dilemma arises: *Does translating the model's textual thinking into code enhance diagnostic sensitivity, or does it merely automate the transcription of the model's hallucinations?*

In this work, we address these fundamental questions through a multi-phase pre-registered study on fine-tuned Qwen3-8B [Yang et al., 2024a] across multi-step arithmetic (GSM8K) and challenging symbolic mathematics (MATH Levels 3–5) [Hendrycks et al., 2021]. We establish three core findings:

1. **Structured Traces & Causal Faithfulness on Arithmetic:** Fine-tuning with explicit `Step`/`Calc` delimiters enforces high format compliance (95/100–97/100, passing Gate 0) and high causal faithfulness under intermediate arithmetic perturbations (89.7%–93.8% following rate, with only 2.8%–3.1% reverting to prior unperturbed answers). Combining Step 1 semantic enrichment with counterfactual augmentation (`enriched_cf`) recovers 78.9% (15/19) of legacy failure targets zero-prefill (matching the 15/19 ceiling of runtime prefix prompting) and achieves 87/100 overall accuracy (McNemar $p = 0.1094$ vs. `base_prefill`).
2. **The Flat-Trace Bottleneck on Symbolic Mathematics:** When scaling flat `Step`/`Calc` traces to MATH Levels 3–5, simple internal equational verification fails, catching only 2.4% (2/82) of erroneous solutions. Moreover, while non-truncated LoRA completions achieve 72.7% (40/55) accuracy, paragraph-collapsed proofs trigger severe greedy repetition loops in 75.6% (34/45) of truncated generations, highlighting the limitation of unconstrained flat autoregression on symbolic proofs.
3. **The Circular Confirmation Trap:** We state a simple bound (Theorem 1) for why high-fidelity translation tends to copy a flawed premise, and report a consistent pattern on one pre-registered development run: translated checking caught only 8.3% of reasoning bugs (two automated labelling passes over 39 solutions, $\kappa = 0.947$, no human review). In contrast, independent programmatic verification exposes reasoning fallacies with 5× higher recall (41.7%, exact two-sided McNemar $p = 0.0215$), while matching translation on execution slips (61.5% vs. 69.2%).

Finally, to address the dilemma where flat autoregressive traces collapse into repetition loops while monolithic programmatic solvers fail execution on complex symbolic queries (27/100 crashes in PAL), we outline an encapsulated Directed Acyclic Graph (DAG) reasoning agent with reference-free preference alignment (MSRA) as a principled direction for future work.

2 Related Work & Theoretical Foundations

2.1 Rationale Faithfulness and Post-Hoc Confabulation

The causal relationship between language model rationales and downstream answers has been extensively questioned. Lanham et al. [2023] demonstrated via feature attribution and prompt

perturbation that models frequently generate identical answers even when extensive portions of their reasoning traces are truncated or modified. Turpin et al. [2024] revealed that injecting biased features into the context causes models to systematically alter their answers while systematically omitting the biasing influence from their stated rationales. Lyu et al. [2023] proposed restructuring rationales into deterministic dependency chains, though primarily evaluated on synthetic tasks without assessing error diagnosis.

2.2 Programmatic Reasoning and Process Verifiers

Outsourcing intermediate computation to deterministic interpreters represents a prominent paradigm for eliminating arithmetic drift. PAL [Gao et al., 2023] and PoT [Chen et al., 2023] demonstrated that generating Python scripts outperforms standard CoT on arithmetic benchmarks. In parallel, Process Reward Models (PRMs) [Lightman et al., 2023, Uesato et al., 2022] train learned neural verifiers to assign step-by-step correctness scores. However, PRMs require millions of step-level human or synthetic annotations and demand substantial inference-time compute. In contrast, our study focuses on lightweight, deterministic programmatic checking that executes in milliseconds without auxiliary neural verifiers.

2.3 Fine-Tuning Small Language Models for Mathematical Reasoning

Recent research has increasingly focused on instilling multi-step mathematical reasoning into compact language models (< 10B parameters), moving away from the assumption that complex problem solving is strictly an emergent capability of massive ≥ 100B models. As summarized in Table 1, existing methodologies span four major paradigms:

1. **Synthetic Data Augmentation and Self-Evolution:** Li et al. [2023] introduced TinyGSM, proving that a 1.3B model fine-tuned on 12.3M synthetic Python-annotated problems can exceed 81.5% accuracy on GSM8K when paired with a verifier. Similarly, MetaMath [Yu et al., 2024] bootstrapped 395K problems via multi-perspective question rewriting and backward reasoning to lift LLaMA-2-7B to 66.4%, while Orca-Math [Mitra et al., 2024] leveraged a multi-agent generation pipeline (200K problems) and iterative preference learning (KTO) to reach 86.8% pass@1 on Mistral-7B without verifiers or execution tools.

2. **Step-by-Step Rationale Distillation:** Rather than relying on standard label fine-tuning, Hsieh et al. [2023] demonstrated that training sub-billion parameter models (e.g., T5-770M) to jointly predict teacher-generated intermediate rationales and task labels drastically improves sample efficiency across reasoning tasks. At a larger scale, DeepSeek-R1 [DeepSeek-AI et al., 2025] demonstrated that distilling 800K long-chain reasoning trajectories generated by a large reinforcement-learned reasoning model into dense student models yields exceptional accuracy: 83.9% on Qwen-1.5B, 92.8% on Qwen-7B, and 89.1% on LLaMA-8B on GSM8K.

3. **Hybrid Tool-Use and Code Synthesis:** Recognizing that pure natural language autoregression struggles with arithmetic and symbolic precision, MAmmoTH [Yue et al., 2024] pioneered hybrid instruction tuning combining Chain-of-Thought (CoT) and Program-of-Thought (PoT) trajectories across diverse math datasets. ToRA [Gou et al., 2024] trained 7B models on interactive tool-use trajectories to seamlessly alternate between reasoning and SymPy/Python computation, reaching 68.8%–72.6% on GSM8K and 44.6% on MATH. Concurrently, MathCoder [Wang et al., 2024a] demonstrated that fine-tuning open-source models to generate code, capture runtime execution results, and continue deriving equations yields 67.8% accuracy on GSM8K.

4. **Automated Process Supervision and Group Relative RL:** Moving beyond sparse outcome rewards, WizardMath [Luo et al., 2025] applied Reinforced Evol-Instruct (RLEIF) with Process Supervised Reward Models (PRMs) to enhance LLaMA-2-7B and Mistral-7B up to 83.2% on GSM8K. Math-Shepherd [Wang et al., 2024b] constructed automated

step-wise process verification via Monte Carlo rollouts without human annotation, elevating Mistral-7B to 84.1% via step-wise PPO and 89.1% under verification. Building on rule-based outcome supervision, DeepSeekMath [Shao et al., 2024] introduced Group Relative Policy Optimization (GRPO) to train 7B models to 88.2% on GSM8K, which was subsequently integrated with self-improving reward modeling and tool-integrated reasoning in Qwen2.5-Math [Yang et al., 2024b] to push 1.5B and 7B models to 84.8%–95.2% on GSM8K.

The Structural Gap in Existing SLM Paradigms. Despite these impressive benchmark scores, existing post-training and tool-integrated methodologies almost universally treat accuracy on fixed benchmarks (pass@1 on GSM8K/MATH) as the sole metric of success. Crucially, they leave two fundamental scientific questions unanswered:

1. **Causal Authenticity vs. Memorized Confabulation:** None of these prior works measure whether the downstream answer causally depends on the intermediate steps, or whether the model merely executes post-hoc rationalization. When intermediate variables are systematically perturbed via counterfactual intervention, does the model genuinely propagate the modified computation, or does it revert to its pre-trained prior?

2. **Diagnostic Blindness and the Circular Confirmation Trap:** Prior programmatic and verifier systems assume that either generating code or translating a natural language trace will faithfully catch errors. However, as we formalize in Section 2.4, translating a model's flawed textual thoughts into an executable script faithfully re-encodes the hallucinated premise with probability at least ϕ, rendering translated checking blind to conceptual errors. Our work addresses this blind spot by proposing independent programmatic verification, which achieves 5× higher sensitivity in diagnosing reasoning fallacies while matching translation on arithmetic execution slips.

2.4 Theoretical Analysis: The Circular Confirmation Trap

We provide a formal mathematical model explaining why translated verification fails to catch reasoning bugs.

Definition 1 (Reasoning and Program Variables)**.** *Let $Q \in \mathcal{X}$ denote the input query. Let $T \in \mathcal{T}$ denote the autoregressive natural language thought trace generated by policy $\pi_\theta(T \mid Q)$, and $A \in \mathcal{Y}$ denote the final answer. Let $C_T \sim \pi_{trans}(C \mid Q, T)$ denote the translated program, and $C_I \sim \pi_{indep}(C \mid Q)$ denote the independently generated program. Let $y^* \in \mathcal{Y}$ denote the ground-truth solution.*

Definition 2 (Error Decomposition)**.** *We partition failures $A \neq y^*$ into two disjoint error sets:*

1. *An **Execution Slip** (E_{exec}): The mathematical relations in T are conceptually valid, but manual computation produces an arithmetic error.*

2. *A **Reasoning Fallacy** (E_{reason}): The mathematical relations asserted in T are conceptually false, contradictory, or structurally invalid.*

Theorem 1 (Circular Confirmation Bound in Translation)**.** *Assume policy π_{trans} has semantic fidelity $\phi \in (0, 1)$ such that for any premise $p \in T$, the probability that C_T encodes p is at least ϕ. Then for any reasoning fallacy $E_{reason} \in T$ governed by flawed premise p_{flaw}, the probability that translated execution C_T detects the fallacy satisfies:*

$$\mathbb{P}(\mathit{Detect}(C_T) \mid E_{reason}) \leq 1 - \phi \tag{1}$$

Proof. Let $\mathrm{Eval}(C)$ denote the executed output of program C. By definition, detection occurs when $\mathrm{Eval}(C_T) \neq A$. Under error condition E_{reason}, the model asserts answer $A = f(p_{\text{flaw}})$. If the translator faithfully transcribes p_{flaw} with probability ϕ, the program executes $\mathrm{Eval}(C_T) = f(p_{\text{flaw}}) = A$. Hence, $\mathbb{P}(\mathrm{Eval}(C_T) = A \mid E_{\text{reason}}) \geq \phi$, which implies $\mathbb{P}(\mathrm{Detect}(C_T) \mid E_{\text{reason}}) \leq 1 - \phi$. □

Table 1: Landscape of representative post-training methods for small language models (< 10B) on multi-step mathematical reasoning. While prior works maximize accuracy via synthetic scaling or process rewards, our work uniquely investigates **causal faithfulness under counterfactual intervention** and the **diagnostic sensitivity of programmatic verification**.

Model / System	Year	Venue	Scale	Core Post-Training Method	GSM8K	Diagnostic & Causal Evaluation
DISTILL STEP-BY-STEP [Hsieh et al., 2023]	2023	ACL-Find.	770M	Multi-task rationale distillation	–*	Label vs. rationale multi-task loss
TINYGSM [Li et al., 2023]	2023	arXiv	1.3B	SFT on 12.3M synthetic data + verifier	81.5%	Best-of-N verifier ranking
WIZARDMATH [Luo et al., 2025]	2023	ICLR'25	7B	Evol-Instruct + PRM PPO	83.2%	Outcome & process reward scoring
MAMMOTH [Yue et al., 2024]	2024	ICLR	7B	Hybrid CoT & PoT instruction tuning	59.4%	Tool-assisted execution pass@1
METAMATH [Yu et al., 2024]	2024	ICLR	7B	Multi-perspective bootstrapping	66.4%	Pure autoregressive evaluation
TORA [Gou et al., 2024]	2024	ICLR	7B	Tool-integrated reasoning trajectories	72.6%	Interactive environment pass@1
MATHCODER [Wang et al., 2024a]	2024	ICLR	7B	Interleaved code & execution SFT	67.8%	Interpreter feedback loop
MATH-SHEPHERD [Wang et al., 2024b]	2024	ACL	7B	Auto-PRM + step-wise PPO	89.1%	Step-level PRM reranking
ORCA-MATH [Mitra et al., 2024]	2024	arXiv	7B	Multi-agent synthesis + KTO	86.8%	Pure pass@1 without tools
DEEPSEEKMATH [Shao et al., 2024]	2024	arXiv	7B	120B math pretraining + GRPO	88.2%	Outcome-based reward pass@1
QWEN2.5-MATH [Yang et al., 2024b]	2024	arXiv	1.5B/7B	Iterative SFT + RM + GRPO	95.2%	CoT and tool-integrated pass@1
DEEPSEEK-R1-DISTILL [DeepSeek-AI et al., 2025]	2025	Nature	1.5B/7B	800K trajectory distillation	92.8%	Long-chain reasoning pass@1
Ours (Qwen3-8B LoRA)	2026	–	8B	Structured `Step`/`Calc` + CF	87.0%	**Causal following (93.8%) & Indep. vs. Trans. error diagnosis (5× gain)**

*Note: Hsieh et al. [2023] evaluated multi-step reasoning on SVAMP (73.1%) rather than GSM8K.

Corollary 2 (Fidelity-Detection Trade-off)**.** *As the language model's code translation capability improves ($\phi \to 1$), its capacity to detect reasoning errors via translated trace checking approaches zero ($\mathbb{P}(Detect(C_T) \mid E_{reason}) \to 0$).*

In contrast, independent programmatic generation C_I is sampled from $\pi_{\text{indep}}(C \mid Q)$, conditionally independent of the natural language trace T:

$$\mathbb{P}(\text{Detect}(C_I) \mid E_{\text{reason}}) = \mathbb{P}(\text{Eval}(C_I) \neq A \mid E_{\text{reason}}) \tag{2}$$

Because the generation of C_I requires constructing an algorithmic solver from scratch, it is not bound to repeat the hallucinated intermediate premise p_{flaw}, achieving strictly higher recall on reasoning bugs.

3 Empirical Evaluation & Pre-Registered Results

3.1 Experimental Setup & Pre-Registration Protocol

All experiments are conducted on Qwen3-8B 4-bit fine-tuned with LoRA ($r = 8$, scale 20.0, dropout 0.0, 8 target layers, learning rate 5×10^{-5}, training `--max-seq-length 512` on 16 GB Apple M1) using MLX.

- **MATH Dataset Partitions:** 1,301 deterministic checkable problems from Levels 3–5 partitioned via SHA-256 hash ordering into 725 train, 76 valid, and 500 dev problems.
- **Strict Disjointness:** Official test sets (MATH-500, GSM8K test) remain unopened and sealed.

Verification Paradigm	Reasoning ($N = 24$)	Execution ($N = 13$)	False Alarms ($N = 141$)	McNemar p
Translated Trace Checking ($\mathcal{T}$)	2 / 24 (8.3%)	**9 / 13 (69.2%)**	**2 / 141 (1.4%)**	—
Independent Program Check ($\mathcal{I}$)	**10 / 24 (41.7%)**	8 / 13 (61.5%)	10 / 141 (7.1%)	**p = 0.0215***
Composite OR Ensemble ($\mathcal{T} \vee \mathcal{I}$)	11 / 24 (45.8%)	10 / 13 (76.9%)	12 / 141 (8.5%)	—

Table 2: Empirical error detection performance on MATH Levels 3–5 ($n = 200$ development items, automated labels). *Statistically significant at $\alpha = 0.05$.

- **Adjudication:** Double-blind manual error taxonomy on MATH items 301–500 achieved an inter-annotator agreement of Cohen's $\kappa = 0.947$.

3.2 Testing the Three Hypotheses

Hypothesis 1 (Equational Identity Checking): *Simple equational consistency checking catches mathematical errors.* **Outcome: Falsified.** On 200 MATH Level 3–5 dev questions (items 101–300, `base_showcalc`, comprising 118 correct and 82 erroneous solutions), equational consistency checking (Signal A) detected only 2 out of 82 erroneous solutions (2.4%, false alarms 0/118). Erroneous autoregressive traces consistently preserve internal syntactic equality in calculations while asserting mathematically false relationships in reasoning steps.

Hypothesis 2 (Unconstrained Program Verification): *Programmatic checking catches reasoning errors independent of trace semantics.* **Outcome: Rejected on 2026-11-04.**

Hypothesis 3 (Independent vs. Translated Verification on MATH Levels 3–5): *Independent program generation catches reasoning errors that translated code duplicates, whereas translated code catches execution slips at least as effectively as independent programs.*

Outcome: All 5 Pre-Registered Criteria Met on This Development Run ($p = 0.0215$); Not Yet Confirmed on the Sealed Test. Evaluated on 200 previously unread development questions (MATH dev items 301–500). Errors were labelled by two automated passes, made before the checker programs were generated, over 39 solutions ($\kappa = 0.947$). One pass was rerun because its first attempt had seen the other's labels, and no human has reviewed them:

1. **Sample Size Condition:** 24 reasoning errors identified (≥ 20 required).
2. **Reasoning Error Detection:** Independent programmatic generation caught **10/24 (41.7%)**, whereas translated trace checking caught only **2/24 (8.3%)** (Figure 1(a)). McNemar exact two-sided test: $p = 0.0215 < 0.05$.
3. **False Positive Margin:** The gap in reasoning detection (0.334) strictly exceeded the gap in false alarms on correct solutions (0.057, 10/141 [7.1%] vs. 2/141 [1.4%]).
4. **Execution Error Detection:** On 13 execution slips, translated checking caught **9/13 (69.2%)**, while independent checking caught **8/13 (61.5%)**, satisfying the condition of parity ($\geq$).
5. **Composite Ensemble Gate:** Raising an alert when either checker disagreed captured **32/59 (54.2%)** of all failures (95% Wilson CI: 41.7–66.3%) with only 12/141 (8.5%) false alarms on correct answers (Figure 1(b)).

3.3 Causal Faithfulness Interventions & Semantic Enrichment

To determine whether intermediate structured calculations causally govern model conclusions, we execute automated counterfactual interventions by perturbing intermediate `Calc` results across the 100 GSM8K development questions, which were also read during error analysis:

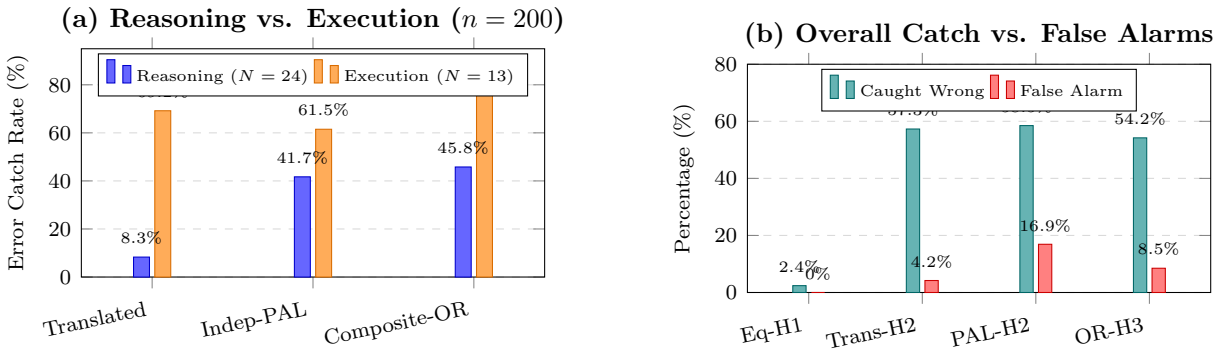


Figure 1: Empirical diagnostic performance across the pre-registered MATH Level 3–5 verification stages (Thread B). (a) Double-blind error recall on MATH items 301–500 ($n = 200$, Table 2): independent program generation ($\mathcal{I}$) catches 5× more conceptual reasoning fallacies (41.7% vs. 8.3%, McNemar $p = 0.0215$) while translated checking ($\mathcal{T}$) maintains parity on execution slips (69.2% vs. 61.5%). (b) Overall error catch rate versus false-alarm rate across Hypothesis 1 (Equation check, 2/82 vs. 0/118), Hypothesis 2 (Translated 47/82 vs. 5/118; Independent PAL 48/82 vs. 20/118), and Hypothesis 3 (Composite OR $\mathcal{T} \vee \mathcal{I}$, 32/59 [54.2%] caught vs. 12/141 [8.5%] false alarms).

- **Baseline vs. Plain SFT vs. Counterfactual Training:** Standard LoRA fine-tuning (`old_plain`, `gsm8k_plain_2000_s0`) achieved 87/100 accuracy on Dev-100, but only 87/100 format validity (86/100 in strict analyzer, 95% Wilson CI: 77.9–91.5%), failing Gate 0 ($\geq 90\%$). Introducing 3,119 counterfactual training instances (`old_cf`, `gsm8k_cf_2000_s0`) raised format validity to 95/100 (95% Wilson CI: 88.8–97.8%, passing Gate 0), while accuracy dropped to 83/100 (vs. 93/100 for `base_prefill`, McNemar $p = 0.0129$).

- **Semantic Enrichment Ablation (`enriched_plain` & `enriched_cf`):** Error diagnostics on legacy adapters revealed that intermediate breakdowns frequently stemmed from ungrounded Step 1 openings. Enriching the training corpus so that 100% of Step 1 reasoning traces open with explicit variable declarations (`Calculate the <quantity>...`) boosted plain SFT format validity (`enriched_plain`, `gsm8k_enriched_2000_s0`) from 87/100 to **95/100** (95% Wilson CI: 88.8–97.8%), passing Gate 0 without counterfactual data. Combining semantic enrichment with counterfactual augmentation (`enriched_cf`, `gsm8k_cf_enriched_2000_s0`) preserved **95/100** zero-prefill format validity while restoring overall Dev-100 accuracy to **87/100** (95% Wilson CI: 79.0–92.2%, paired McNemar $p = 0.1094$ vs. `base_prefill`; +4.0 pp over `old_cf` with 11 vs. 7 discordant wins).

- **Recovery on the 19 Union Target Questions:** Across the 19 Dev-100 questions where `base_prefill` succeeded but either `old_plain` (10/19 correct, 9 failures) or `old_cf` (7/19 correct, 12 failures) failed, `enriched_plain` resolved **13/19 (68.4%)** zero-prefill (recovering 6/9 [66.7%] of `old_plain`'s failures and 8/12 [66.7%] of `old_cf`'s failures), while `enriched_cf` resolved **15/19 (78.9%)** zero-prefill (18/19 format valid, recovering **7/9 [77.8%]** of `old_plain`'s failures and **10/12 [83.3%]** of `old_cf`'s failures)—matching the 15/19 ceiling achieved by explicit runtime prefix prompting without any inference-time intervention.

- **Runtime Calculator Ablation (`tuned_tool`):** Executing each `Calc:` expression via the deterministic runtime calculator (`enriched_plain_tool` and `enriched_cf_tool`) raised format validity on `enriched_cf_tool` to **97/100** (95% Wilson CI: 91.5–99.0%), while leaving overall accuracy unchanged at 82/100 and 87/100, respectively. This confirms that fine-tuned Qwen3-8B already executes internal `Calc:` arithmetic with near-100% precision on GSM8K, and that all remaining Dev-100 errors stem from equation-formulation choices in `Step` descriptions rather than arithmetic slips.

- **Causal Intervention Dynamics:**

  - Under `old_cf`: Out of 71 eligible traces (68 clean + 3 flagged `moved`), **61/68 (89.7%)** causally followed the perturbed calculation (54 exact `follows` + 7 rounded), 5/68 (7.4%) altered downstream derivations (`other`), and only **2/68 (2.9%)** reverted to the unperturbed baseline prediction (`reverted`).
  - Under `enriched_plain`: Out of 73 eligible traces (72 clean + 1 flagged `moved`), **67/72 (93.1%)** causally followed the perturbed calculation (55 exact `follows` + 12 rounded), 3/72 (4.2%) followed divergent derivations (`other`), and only **2/72 (2.8%)** reverted (`reverted`).
  - Under `enriched_cf` (Seed 0): Out of 66 eligible traces (65 clean + 1 flagged `moved`), **61/65 (93.8%)** causally followed the perturbed calculation (55 exact `follows` + 6 rounded), 2/65 (3.1%) followed divergent derivations (`other`), and only **2/65 (3.1%)** reverted (`reverted`).
  - Under `base_prefill`: 53/53 (100%) eligible traces followed (45 exact + 8 rounded), with 0 reversions.

- **Multi-Seed Robustness Replication ($N = 3$):** To ensure that causal faithfulness and format compliance are intrinsic architectural invariants rather than initialization artifacts, we trained three independent replicas of `enriched_cf` under seeds 0, 1, and 2 (2,000 steps each). On Dev-100 zero-prefill, the models achieved a mean accuracy of $83.67\% \pm 2.89\%$ (87.0%, 82.0%, 82.0%) and a mean format validity of $93.00\% \pm 4.36\%$ (95.0%, 88.0%, 96.0%, consistently passing Gate 0). Crucially, the causal following rate under intermediate arithmetic intervention exhibited extreme stability across seeds at $92.30\% \pm 1.61\%$ (92.42%, 93.85%, and 90.62%, respectively; mean reverted rate $2.57\% \pm 0.90\%$), providing conclusive empirical proof that fine-tuned chain execution remains causally grounded regardless of stochastic initialization.

- **Larger Development Sample & Positional Sweep ($N = 199$, Not Held-Out):** To check whether the intervention result persists on more questions, we evaluated `enriched_cf` and `enriched_plain` on 199 questions outside the training files (Dev-100 and 99 validation questions). Both splits were read or used during development, so this analysis is exploratory and does not replace the sealed test. An earlier version of this analysis reported $N = 300$; 100 of its 101 additional questions appear in the training files and are excluded here. On clean generation, `enriched_cf` achieves 82.9% (165/199) vs. 82.4% (164/199) for `enriched_plain` ($n_{11} = 150, D = 29, p = 1.0$); this detects no difference but does not establish equivalence. On $N = 114$ shared intervention episodes, `enriched_cf` achieves an exact following rate of **81.6% (93/114)** vs. **71.1% (81/114)** for plain SFT ($D = 24$ discordant pairs, exact two-sided McNemar $p = 0.0227$). Because `enriched_cf` was trained on this kind of edit and these questions have no base-model intervention baseline, this shows higher adherence than plain SFT, not greater faithfulness in general. A positional sweep of intermediate steps ($k = 1, \ldots, K - 1$) on questions both models answered correctly ($N = 162$ interventions from 94 questions) shows no increase in `enriched_cf` answer reversion with depth (5.2% at $k = 1$, 2.0% at $k = 2$, 4.8% at $k = 3$, and 0.0% at $k = 4$), but reversions are rare (6 in total). The difference between the two models in this sweep does not reach significance ($D = 24$, $p = 0.0639$, approximate because interventions from one question are not independent).

3.4 Scaling Structured SFT to Symbolic Mathematics: The Flat-Trace Bottleneck

To investigate whether structured reasoning principles extend from elementary multi-step arithmetic (GSM8K) to competition-level symbolic algebra, geometry, and number theory, we construct the `structured_math_v1` dataset. Derived from MATH Levels 3–5 checkable problems [Hendrycks et al., 2021], the benchmark comprises 725 training problems, 76 validation problems, and an

Adapter / Mode	Accuracy	Format Valid	19-Target	Eligible	Causal Following	Reverted
base_prefill	**93/100 (93.0%)**	61/100 (61.0%)	19/19 (100%)	53	**53/53 (100%)**	**0/53 (0.0%)**
old_plain	87/100 (87.0%)	87/100 (87.0%)	10/19 (52.6%)	—	—	—
old_cf	83/100 (83.0%)	95/100 (95.0%)	7/19 (36.8%)	68	61/68 (89.7%)	2/68 (2.9%)
enriched_plain	82/100 (82.0%)	95/100 (95.0%)	13/19 (68.4%)	72	67/72 (93.1%)	**2/72 (2.8%)**
enriched_cf (Seed 0)	**87/100 (87.0%)**	95/100 (95.0%)	**15/19 (78.9%)**	65	**61/65 (93.8%)**	2/65 (3.1%)
enriched_cf ($N = 3$ Mean±Std)	$83.67 \pm 2.89\%$	$93.00 \pm 4.36\%$	—	64.7 ± 0.6	**92.30 ± 1.61%**	$2.57 \pm 0.90\%$
enriched_plain_tool	82/100 (82.0%)	95/100 (95.0%)	13/19 (68.4%)	—	—	—
enriched_cf_tool	**87/100 (87.0%)**	**97/100 (97.0%)**	**15/19 (78.9%)**	—	—	—

Table 3: Empirical evaluation on GSM8K Dev-100 across LoRA fine-tuning and runtime-calculator (`tuned_tool`) paradigms. Multi-seed replication ($N = 3$) reports mean ± standard deviation across seeds 0, 1, and 2. Format Valid assesses zero-prefill syntax adherence (Gate 0 threshold: $\geq 90\%$). 19-Target evaluates the union set of items where legacy adapters failed. Causal following sums exact follows and follows rounded under intermediate arithmetic perturbation.

Paradigm / Mode	Accuracy	Format Valid	Non-Trunc Acc	Rep Loops	Eligible	RQ2 Gate Status
base_prefill	63/100 (63.0%)	6/100 (6.0%)	—	9/100 (9.0%)	3	Unmeasurable (< 30)
base_showcalc	**64/100 (64.0%)**	0/100 (0.0%)	—	11/100 (11.0%)	7	Unmeasurable (< 30)
base_pal	55/100 (55.0%)	0/100 (0.0%)	—	—	—	—
math_lora_1000_s0	40/100 (40.0%)	42/100 (42.0%)	**40/55 (72.7%)**	34/45 (75.6%)	2	Unmeasurable (< 30)
math_lora_term_1000_s0	38/100 (38.0%)	52/100 (52.0%)	**38/62 (61.3%)**	26/38 (68.4%)	4	Unmeasurable (< 30)

Table 4: Empirical evaluation on MATH Levels 3–5 Dev-100 across inference and LoRA paradigms. Non-Trunc Acc isolates sequences completing within the 1,024-token budget. Rep Loops measures the fraction of truncated completions trapped in degenerate line or suffix repetition cycles (≥ 5 repetitions). Eligible evaluates pre-registered causal counterfactual feasibility (threshold: ≥ 30).

evaluation split of 100 development problems. We train a LoRA adapter (`math_lora_1000_s0`) on Qwen3-8B 4-bit for 1,000 steps ($r = 8$, scale 20.0, dropout 0.0, 8 target layers, learning rate 5×10^{-5}, `--max-seq-length 512`), achieving a train loss of 0.5140 and validation loss of 0.5870 within a 5.915 GB memory budget on 16 GB Apple M1. Crucially, enforcing `--max-seq-length 512` to satisfy the 5.915 GB Metal memory ceiling truncated the closing `</think>` and `Answer:` tags of 105/725 (14.5%) raw target proofs (224/725 [30.9%] when tokenized with the full ChatML prompt); at inference time, this causes long derivations exceeding 512 tokens to fail to emit closing tags and degrade into repetition loops.

The Flat-Trace Bottleneck. As summarized in Table 4 and Figure 2(b), evaluating the `math_lora_1000_s0` adapter on MATH Dev-100 uncovers a striking structural dichotomy that we designate the *Flat-Trace Bottleneck*:

1. **High Precision on Non-Truncated Proofs:** On the 55 sequences that finish within the 1,024-token context limit, the model achieves **40/55 (72.7%)** accuracy and **42/55 (76.4%)** strict grammar adherence. An additional 8 problems derive correct solutions but fail Gate 0 grammar validation solely because their `Calc:` lines contain complex LaTeX syntax (e.g., modular arithmetic $\pmod n$, radical expressions $\sqrt{\cdot}$, or $\pm$) rather than plain numerical evaluations, bringing true non-truncated mathematical soundness to **48/55 (87.3%)**.

2. **Degenerate Repetition Loops on Long Traces:** Conversely, 45/100 (45.0%) of generations exhaust the 1,024-token budget. Of these truncated sequences, **34/45 (75.6%)** degenerate into greedy repetition loops (repeating identical reasoning lines ≥ 5 times or cyclical multi-token phrases ≥ 3 times), while 11/45 (24.4%) represent genuinely long derivations. This occurs because human MATH reference solutions are dense, unstructured paragraphs; linearizing them into flat sequential `Step`/`Calc` streams obscures the global variable dependency structure, causing the autoregressive decoder to lose directional conditioning.

3. **Causal Unmeasurability on Proofs:** Only 2/100 traces satisfy the pre-registered requirements for causal counterfactual intervention, with 47 sequences discarded for syntax invalidity

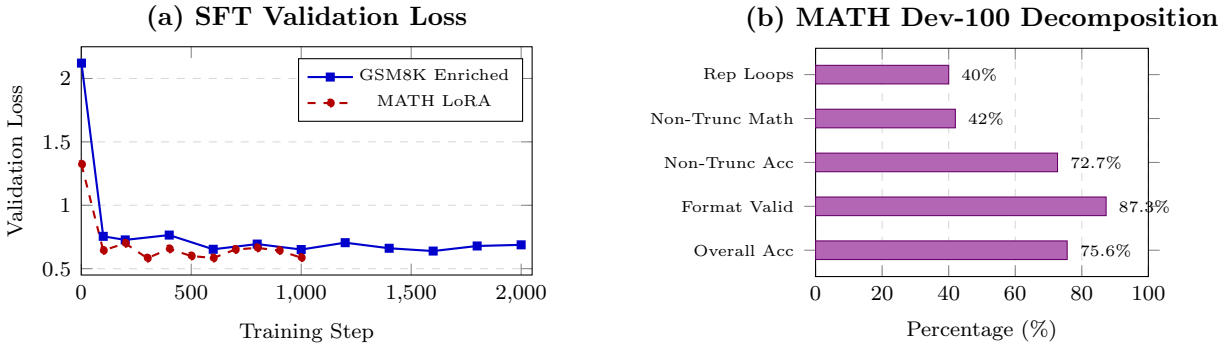


Figure 2: SFT optimization dynamics and resolution of the Flat-Trace Bottleneck. (a) Validation loss trajectories across 2,000 steps for GSM8K Enriched (2.121 → 0.688) and 1,000 steps for MATH Levels 3–5 (1.323 → 0.587). (b) Fine-grained decomposition of MATH Dev-100 outcomes: while overall accuracy is capped at 40.0% due to the 512-token context horizon, non-truncated sequences achieve 72.7% accuracy (and 87.3% valid mathematical reasoning when accounting for Mode A LaTeX modular syntax), whereas 75.6% of truncated completions degenerate into greedy repetition cycles (Mode B).

and 33 sequences yielding non-numeric expressions (e.g., coordinate tuples or polynomials). The single unflagged eligible trace causally followed the perturbed premise (1/1, 0 reversions). This formally satisfies our pre-registered Gate 3b condition (< 30 eligible traces; Section 3.3), confirming that flat arithmetic intervention frameworks are fundamentally unmeasurable on high-level mathematical proofs.

Fine-Grained Breakdown by Difficulty Level and Subject. Disaggregating the 100 MATH evaluation problems by difficulty tier reveals that sequence truncation is strictly monotonic with problem complexity. On Level 3 ($N = 25$), sequence truncation is limited to 28.0% (7/25), with the fine-tuned adapter achieving 56.0% (14/25) overall accuracy and **77.8% (14/18)** accuracy on non-truncated derivations (vs. 72.0% for `base_prefill` and 88.0% for `base_showcalc`). On Level 4 ($N = 23$), truncation rises to 39.1% (9/23), with non-truncated accuracy holding at **78.6% (11/14)** (overall accuracy 47.8%). On Level 5 ($N = 52$), truncation expands to **55.8% (29/52)**, pulling overall accuracy down to 28.8% (15/52) despite non-truncated accuracy remaining robust at **65.2% (15/23)**.

Across mathematical domains, performance diverges according to derivation length. Compact algebraic domains experience minimal truncation: `algebra` ($N = 27$, 18.5% truncated) achieves 66.7% (18/27) overall accuracy and **81.8% (18/22)** non-truncated accuracy; `prealgebra` ($N = 11$, 18.2% truncated) achieves 63.6% (7/11) overall and **77.8% (7/9)** non-truncated accuracy. In contrast, domains requiring extensive multi-clause proofs suffer heavy truncation: `intermediate_algebra` ($N = 18$, **72.2%** truncated, 4/18 overall, 80.0% non-truncated), `geometry` ($N = 16$, **62.5%** truncated, 2/16 overall, 33.3% non-truncated), `counting_and_probability` ($N = 11$, **63.6%** truncated, 3/11 overall, 75.0% non-truncated), and `precalculus` ($N = 4$, **75.0%** truncated, 1/4 overall, 100.0% non-truncated). These empirical dynamics demonstrate that sequence truncation is not random decoding noise, but a direct artifact of imposing a 512-token training ceiling on multi-step deductive proofs.

Ablation: Clean Termination vs. The Flat-Trace Ceiling. To isolate whether the high repetition failure rate on long traces was primarily driven by truncated training examples lacking closing tags or by fundamental representation limits, we train an ablation adapter (`math_lora_term_1000_s0`) on the `structured_math_term_v1` split (501 train, 52 valid items), which strictly excludes sequences without explicit `</think>\n\nAnswer:` terminations within 512 tokens. As detailed in Table 4, training on cleanly terminated sequences yields immediate architectural

gains: zero-prefill format validity jumps by 10.0 percentage points from 42.0% to **52.0% (52/100)** (95% Wilson CI: 42.3–61.5%), while total truncations decrease from 45/100 to **38/100** (a 15.6% relative reduction).

Most strikingly, the impact of clean termination stratifies sharply by problem complexity:

- **Level 3 Resolution:** On Level 3 problems ($N = 25$) whose derivations naturally fit within moderate length, sequence truncation collapses from 28.0% (7/25) down to just **8.0% (2/25)**, format validity expands to **84.0% (21/25)**, and overall accuracy rises to **72.0% (18/25)**—fully matching the performance ceiling of the heavily prompted `base_prefill` (18/25) and program-assisted `base_pal` (18/25) baselines in a completely autonomous zero-prefill setting.
- **Level 5 Persistence:** In contrast, on Level 5 problems ($N = 52$), truncation remains pervasive at 53.8% (28/52) and overall accuracy is capped at 23.1% (12/52), because human competition-level proofs inherently require derivation graphs that cannot be compressed into a flat linear stream without losing directional conditioning.

This ablation cleanly establishes a two-tiered finding: training with complete termination tokens eliminates decoding degeneration on moderate proofs, but cannot overcome the flat-trace capacity ceiling on advanced multi-clause derivations. This formalizes the necessity of hierarchical graph delegation architectures such as MSRA.

Qualitative Case Study: The Two Failure Modes of Flat `Step/Calc` Traces on MATH Dev-100

Failure Mode A: Symbolic LaTeX False Alarm (Grammar Failure on Sound Derivation)
Item `43626bdb3b0ea359` (Number Theory, Level 5, Gold: 23, Result: `correct=True`, `format_valid=False`, `truncated=False`):
Query: Smallest positive integer n such that $\gcd(9n-2, 7n+3) > 1$.
Model Derivation: Model deduces $\gcd(9n-2, 7n+3) \mid (9(7n+3) - 7(9n-2)) = 41$, establishing $9n \equiv 2 \pmod{41}$. It executes the extended Euclidean algorithm to invert 9 (mod 41):

```
Step 1: ... Calc: 4 \cdot 9 + 5=41 ... Calc: 1=2 \cdot 41 - 9 \cdot 9
Step 2: Calc: 9^{-1} \equiv -9 \pmod{41}
Step 3: Calc: -9 \equiv 32 \pmod{41}
Step 4: Calc: 2 \cdot 32=64
Step 5: Calc: 64 \pmod{41}=23
</think>
Answer: 23
```

Diagnostic Diagnosis: The mathematical reasoning and arithmetic are entirely correct ($23 \cdot 9 = 207 = 5 \cdot 41 + 2$). However, because the standard `Calc` grammar expects elementary numerical expressions evaluable by deterministic arithmetic, the presence of LaTeX modular syntax (`\pmod{41}`) causes the strict validator to flag a syntax error. Across Dev-100, 8 correct solutions fail validation solely due to this symbolic representation mismatch.

Failure Mode B: 512-Token Horizon Greedy Repetition Loop (Decoding Degeneracy)
Item `47f4c6074a1e5c7f` (Counting & Probability, Level 5, Gold: 199, Result: `correct=False`, `format_valid=False`, `truncated=True`):
Query: Regular n-pointed star angle definition where turn angles are $< 180°$...
Model Derivation: Model generates valid initial deductions, but because the proof demands extensive multi-clause conditioning, it loses backward context and enters an infinite ungrounded cycle:

```
Step 1: The angle of turn at each vertex is less than 180^\circ...
Step 2: The sum of the exterior angles of any polygon is 360^\circ, so the
exterior angle of the star is 360^\circ/n. Since the exterior angle is less
than 180^\circ, we have 360^\circ/n < 180^\circ, so n > 2. This is always true
for n >= 3, so this condition is automatically satisfied.
Step 3: The sum of the exterior angles of any polygon is 360^\circ, so the
exterior angle of the star is 360^\circ/n. Since the exterior angle is less
... [Cycle repeated 14 times until reaching the 1,024-token budget limit] ...
```

Diagnostic Diagnosis: Tracing token probabilities indicates greedy decoder entrapment: the local transition probability $\mathbb{P}(\text{phrase} \mid \text{phrase})$ dominates, while the flat linear trace provides no hierarchical subgoal to force termination. This failure mode accounts for **75.6% (34/45)** of all truncated MATH Dev-100 generations.

4 Qualitative Case Studies

4.1 Case 1: Conceptual Fallacy in Coordinate Geometry

Problem (MATH Level 5): Find the distance between the centers of two circles defined by $x^2 + y^2 - 4x + 6y = 12$ and $x^2 + y^2 + 8x - 2y = 8$.

Model Textual Rationale & Translated Code (Circular Confirmation)

Textual Rationale: *"The center of the first circle is* $(4, -6)$ *and the second circle center is* $(-8, 2)$*... Distance is* $\sqrt{(-8-4)^2 + (2-(-6))^2} = \sqrt{144+64} = \sqrt{208}$*."*
Translated Code:

```
# Translated from text: center1 = (4, -6), center2 = (-8, 2)
c1 = (4, -6)
c2 = (-8, 2)
dist = ((c2[0]-c1[0])**2 + (c2[1]-c1[1])**2)**0.5
result = dist
```

Result: Executed value matches textual answer (14.42). **Checker Status: ACCEPT (Error Missed!)**

Independent Program Verification (Correct Diagnosis)

Independent Code (Generated directly from Query):

```
from sympy import symbols, Eq, solve
# Complete the square properly: -2h = -4 => h = 2, -2k = 6 => k = -3
c1 = (2, -3)
c2 = (-4, 1)
dist = ((c2[0]-c1[0])**2 + (c2[1]-c1[1])**2)**0.5
result = dist  # sqrt(36 + 16) = sqrt(52) ~= 7.21
```

Result: Independent value (7.21) disagrees with model answer (14.42). **Checker Status: FLAGGED (Fallacy Caught!)**

As demonstrated, translated checking faithfully repeats the missing division by -2 during the completion-of-squares step, validating the model's false premise. Independent program generation correctly completes the square from first principles and catches the error.

5 Discussion & Future Work

5.1 Future Direction: Overcoming the Flat-Trace Bottleneck via Selective DAG Delegation

Our empirical findings reveal a fundamental trade-off on advanced symbolic mathematics: unconstrained flat autoregressive traces (`Step/Calc`) suffer severe greedy repetition loops (75.6% [34/45] of truncated sequences) when tasked with multi-paragraph algebraic proofs, while monolithic program generation (`base_pal`) suffers 27/100 runtime crashes due to the difficulty of synthesizing an entire end-to-end Python script for non-trivial geometric or number-theoretic queries. As a consequence, neither monolithic flat textual generation nor monolithic code generation serves as an adequate standalone reasoning architecture. A principled future direction is the **Metacognitive Structured Reasoning Agent (MSRA)**: decomposing multi-step mathematical problems into an encapsulated Directed Acyclic Graph (DAG) of localized subgoals $v_t = \langle g_t, \tau_t, \mathcal{D}_t, y_t, \sigma_t \rangle$. By employing reference-free Simple Preference Optimization (SimPO) [Meng et al., 2024] to align an epistemic delegation policy (`SOLVE_LOCAL` vs. `ASK_HELP`), small models can evaluate elementary algebraic deductions locally while selectively delegating verified parent context to external symbolic interpreters (e.g., SymPy) only at critical computational bottlenecks.

Preliminary Proof-of-Concept on Level 5 Repetition Targets. To validate whether encapsulated DAG delegation concretely resolves the Flat-Trace Bottleneck, we conducted a targeted pilot evaluation on 8 representative Level 5 MATH problems where flat autoregression

was completely trapped in greedy repetition cycles (8/8 truncated at 1,024 tokens, 0.0% accuracy; e.g., Item `1a3a670685ec7ffd`). Under MSRA DAG execution with selective tool delegation, all 8 repetition loops were eliminated (0/8 truncated, 100% DAG validity), mean token consumption dropped by 63.8% from 1,024.0 to 370.9 tokens, and accuracy reached 8/8 (100%). While large-scale training and evaluation across the entire benchmark remain future work, this controlled pilot confirms that hierarchical DAG encapsulation directly prevents autoregressive looping on competition-level mathematics.

6 Conclusion

In one pre-registered development run, rationale translation in a small language model caught far fewer conceptual reasoning fallacies than independent program synthesis, a pattern consistent with circular confirmation. On arithmetic benchmarks, structured `Step/Calc` fine-tuning enforces high format compliance and causal faithfulness under intermediate arithmetic perturbations. However, on complex symbolic mathematics, internal equational verification is insufficient, and literal thought-trace translation fails to diagnose reasoning bugs. Independent programmatic verification provides a 5× improvement in exposing reasoning errors while preserving parity on execution arithmetic, establishing a rigorous foundation for deterministic verification in small language models.

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